

## Exp. No: 5

**Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.**

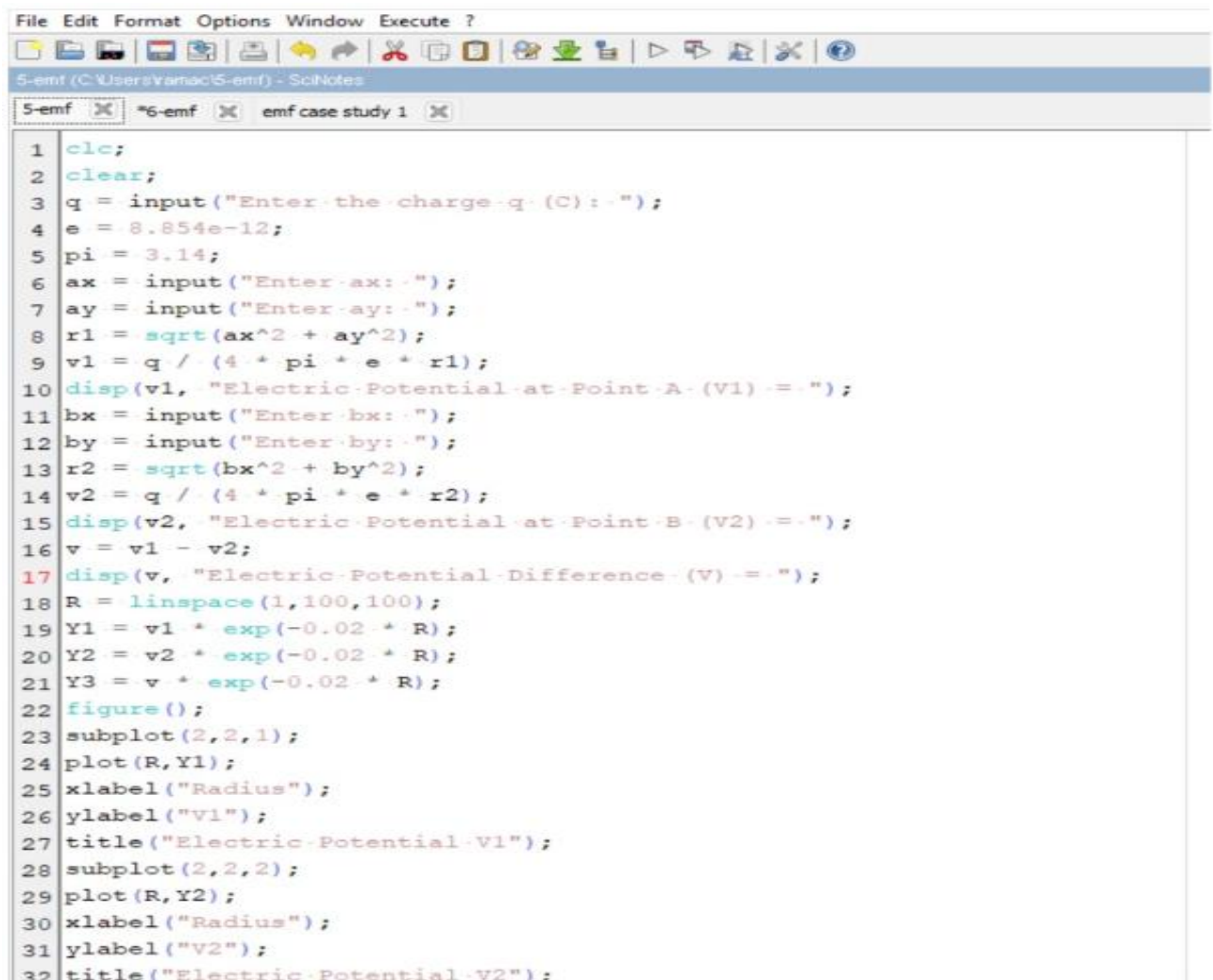
### Aim:

To write program for Electric potential difference between two points in free space medium using SCILAB.

### Software required:

- SCILAB version 6.0.1

### Partical calucation:



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5-emf (C:\Users\Varad\5-emf) - SciNotes
5-emf X *6-emf X emf case study 1 X

1 clc;
2 clear;
3 q = input("Enter the charge q (C) : ");
4 e = 8.854e-12;
5 pi = 3.14;
6 ax = input("Enter ax: ");
7 ay = input("Enter ay: ");
8 r1 = sqrt(ax^2 + ay^2);
9 v1 = q / (4 * pi * e * r1);
10 disp(v1, "Electric Potential at Point A (V1) :-");
11 bx = input("Enter bx: ");
12 by = input("Enter by: ");
13 r2 = sqrt(bx^2 + by^2);
14 v2 = q / (4 * pi * e * r2);
15 disp(v2, "Electric Potential at Point B (V2) :-");
16 v = v1 - v2;
17 disp(v, "Electric Potential Difference (V) :-");
18 R = linspace(1,100,100);
19 Y1 = v1 * exp(-0.02 * R);
20 Y2 = v2 * exp(-0.02 * R);
21 Y3 = v * exp(-0.02 * R);
22 figure();
23 subplot(2,2,1);
24 plot(R,Y1);
25 xlabel("Radius");
26 ylabel("V1");
27 title("Electric Potential V1");
28 subplot(2,2,2);
29 plot(R,Y2);
30 xlabel("Radius");
31 ylabel("V2");
32 title("Electric Potential V2");
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File Edit Control Applications ?
Scilab 6.1.1 Console

Enter the charge q (C): 2*10^-6

Enter ax: 0.03

Enter ay: 0.04

359692.05

"Electric Potential at Point A (V1) = "
Enter bx: 0.02

Enter by: 0.05

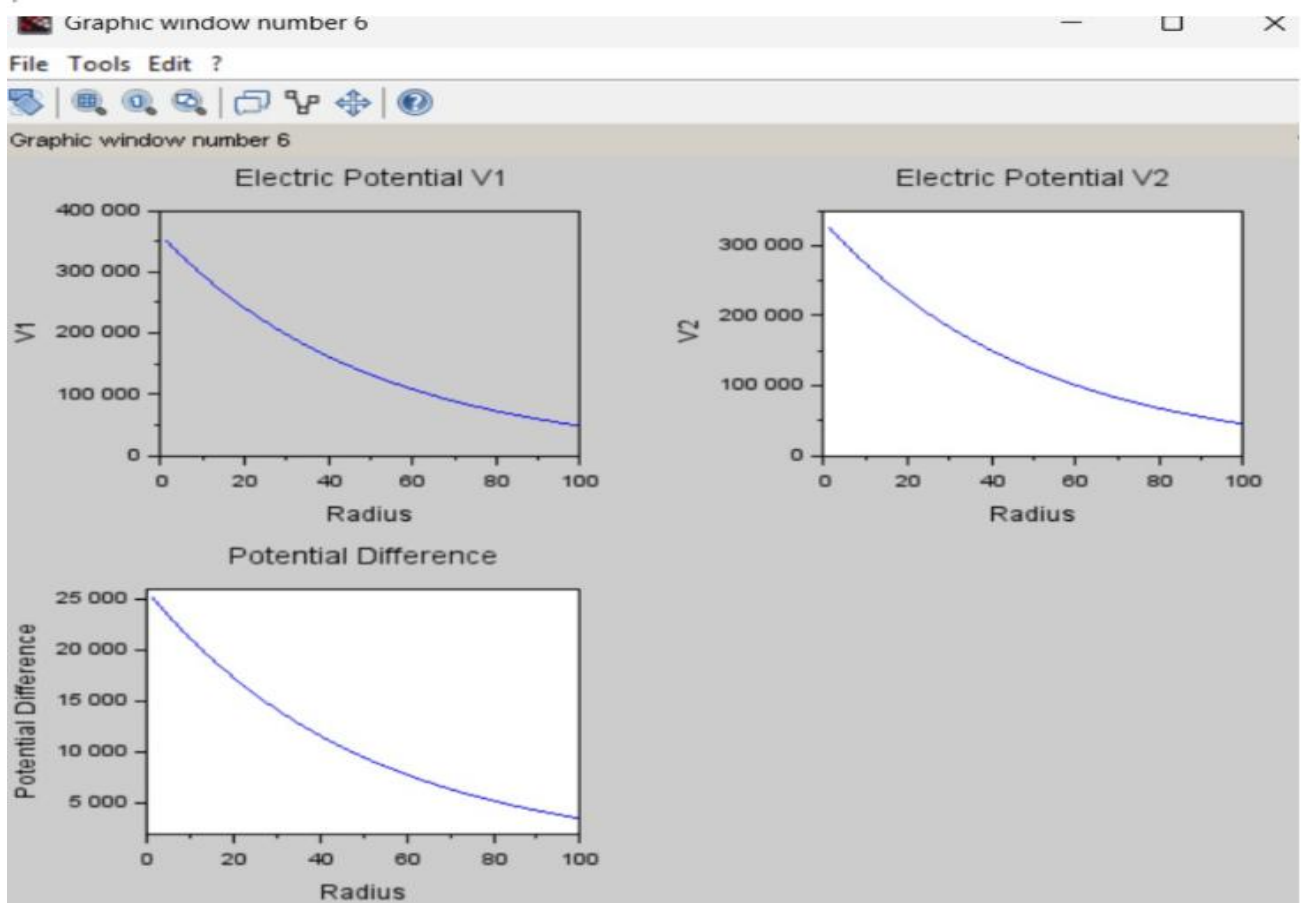
333965.68

"Electric Potential at Point B (V2) = "

25726.365

"Electric Potential Difference (V) = "

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### Theoretical calculation:

Handwritten calculations on a piece of paper:

1)  $a_x = 3$   
 $b_x = 2$   
 $a_y = 4$   
 $b_y = 5$   
 $q = 2 \times 10^{-6}$   
 $r = 3 \text{ mm}$

2)  $r_1 = \sqrt{a_x^2 + b_x^2}$   
 $= \sqrt{(0.03)^2 + (0.02)^2}$   
 $= 3.59692 \times 10^{-2}$

3)  $V_1 = \frac{q}{4\pi\epsilon_0 r_1} = \frac{2 \times 10^{-6}}{4(8.85 \times 10^{-12})(0.0223)}$   
 $= 0.02$

4)  $r_2 = \sqrt{a_y^2 + b_y^2}$   
 $= \sqrt{(0.04)^2 + (0.05)^2}$   
 $= 3.83965 \times 10^{-2}$

5)  $V_2 = \frac{q}{4\pi\epsilon_0 r_2} = \frac{2 \times 10^{-6}}{4(8.85 \times 10^{-12})(0.0383965)}$   
 $= 25726$

### RESULT:

The electric potential difference for different values of was successfully calculated and plotted.